applied sciences

MDPI

Article

Embrace the Era of Drones: A New Practical Design Approach to Emergency Rescue Drones

Zhiyuan Wang [1], Ke Yang [2,*], Yonggang Wang [1], Zechen Zhu [1] and Xiuli Liang [1]

[1] School of Mechanical and Electrical Engineering, Anhui University of Science and Technology, Huainan 232001, China; xinyuan8610@126.com (Z.W.)
[2] School of Carbon Neutralily Science and Engineering, Anhui University of Science and Technology, Hefei 230000, China
* Correspondence: 2023183@aust.edu.cn

Abstract: To increase user satisfaction with emergency rescue drone products, a product modelling design method based on the fuzzy Kano-QFD-FBS model is proposed. First, the initial user requirements for the emergency rescue drone products are obtained through a questionnaire, and the fuzzy Kano model is utilised and combined with the better–worse coefficient method to categorise the attributes, define the priorities of the user requirements, and screen out the key user requirements. Second, the QFD model is used to construct the quality house, analyse the key user requirements quantitatively, and obtain the design elements and weights of the emergency rescue drone product. The obtained key design elements are subsequently imported into the FBS model to complete the mapping transformation from the functional elements to the structural elements of the emergency rescue drone products and realise the styling design of the emergency rescue drone products. Finally, the user satisfaction scale based on appearance, functionality, and interaction was developed and the System Usability Scale (SUS) was used to evaluate user satisfaction with the emergency rescue drone design scheme. The new design scheme scored higher and showed significant differences in satisfaction ratings compared to the previous scheme. Hefei Jiaxun Technology Co., Ltd. carried out product development for the design scheme. At present, physical products have been sold on the market and have achieved good results. Hefei Jiaxun Technology Co., Ltd. conducted a survey on consumer satisfaction with this product, and the results revealed that customer satisfaction increased by 11.9% compared with that of previous products. Compared with similar products in the market, the consumer satisfaction with this product increased by 13.5%, indicating that it has obvious market competitiveness. This study shows that the method of product styling design based on the fuzzy Kano-QFD-FBS model can comprehensively acquire and analyse user requirements, realise accurate mapping from user requirements to product design elements, and output the specific solution of the emergency rescue drone product styling design. The design scheme performs well in meeting user requirements, verifies the feasibility and effectiveness of the fuzzy Kano-QFD-FBS model in the styling design study of emergency rescue drones, and provides a new paradigm for emergency rescue product design.

Keywords: user requirements; fuzzy Kano; QFD; FBS; emergency rescue drones

Academic Editor: Kambiz Vafai

Received: 14 November 2024
Revised: 18 December 2024
Accepted: 19 December 2024
Published: 27 December 2024

Citation: Wang, Z.; Yang, K.; Wang, Y.; Zhu, Z.; Liang, X. Embrace the Era of Drones: A New Practical Design Approach to Emergency Rescue Drones. *Appl. Sci.* **2025**, *15*, 135. https://doi.org/10.3390/app15010135

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

At present, with the frequent occurrence of various types of disasters, China has attached increasing importance to emergency rescue work. According to the content of the '14th Five-Year Plan' National Emergency System Planning Notice issued by the State Council, China is one of the countries with the most severe natural disasters in the world. The emergency industry and emergency equipment provide important technical support for emergency management work through the state's strong support and encouragement as stated in 'The 13th Five-Year Plan for the Development of National Emergency Response System'. With the development of high-level and new technology, aerial vehicle technology has become a new fire rescue method, especially emergency rescue drones, because of their high efficiency, speed, and safety characteristics. This technology has received increasing attention and has been increasingly used by fire departments [1]. In the rescue process, drones effectively safeguard people's lives and minimise losses through real-time monitoring, rapid positioning, communication relays, and other functions. The application of drones from standby has increasingly become necessary and is becoming an important tool for modern urban emergency rescue.

Research on drones started relatively late in China. It was only in the 1970s that China began to develop its own reconnaissance aircraft, the scouting plane, which at that time was mainly used for military and target aircraft training. Since the 1990s, China's drones have made remarkable progress in technology, and their application scope has expanded from the military to communications, elephants, and other fields. The typical research conducted on these machines is summarised as follows. Nan [2] proposed that to integrate multiple function modules, such as body analysis, image acquisition, tilt photography, infrared temperature measurement, bright light illumination, and an extinguishing device, improving the specialisation, integration, and intelligent levelling of drones is necessary. Cheng [3] analysed the technical advantages of the drone kill control system when he explored the control system and key technology of drone kill to clarify the function and performance control system requirements and explored key technologies such as HD video transmission, video decoding, GIS, and online communication. Wang et al. [4] proposed a fast and accurate identification method for flame marks based on the Kalman filtering algorithm and mean shift in a visual algorithm for the destruction of a building's floor to solve the problem of signal loss or misjudgement caused by "ruler" changes or blockage.

From the perspective of the global market, the United States, East Asia, and Europe constitute the main volume of the global market for industrial-grade drones. The United States leads the world and accounts for approximately 33% of the market share. In 2018, the global market for industrial-grade machines reached USD 3.7 billion and rapidly increased to USD 4.8 billion in 2019, a significant increase of approximately 30.7%. Research on and response to emergency rescue drones abroad has achieved positive results. For example, the Indago drone is equipped with an advanced Kestrel mobile driving system, and its main function is intelligent disaster detection. The key feature of the drone is that the fault repair algorithm is adopted, and the amplitude improves the security and reliability of the drone. During the volcanic eruptions in Australia in 2020, the country's rescue team took advantage of the measurement of disaster volume and early warning of disasters. Drones have unique advantages in preventing and extinguishing the threat of disasters. They can reduce the difficulty, cost, and risk coefficient of the operation and simultaneously improve the efficiency and quality of the operation to effectively reduce the casualty insurance and economic loss caused by the disaster. In the past few years, remarkable progress has been made in the research of aviation machine technology, which covers

structural design, operation style, performance, and navigation control systems, especially aircraft types. It can be applied to multiple domains and effectively improves the efficiency and safety of operation [5].

Previous studies on drones have focused mainly on technical systems [6,7], hardware fusion [8], structural science [9], and research methods [10,11]. These studies have analysed different response types and different research degrees, but there is a lack of research on drone modelling [12,13] and a lack of integration of innovative models in the process of product development and design path expansion [14]. The development and evolution of drone modelling is an evolutionary process, and its characteristics are influenced by functional requirements, technical conditions, cognitive balance, and artistic aesthetics in different periods. Drone modelling has experienced three stages: the guiding period of functionalism, the development period of dynamic science, and the fusion period of industrial design. The evolution of modelling reflects the gradual transformation of functional value to aesthetic value. In the fusion period of industrial design, the shape of the drone presents the development trend of diversification, miniaturisation, block moulding, multirotor incorporation, and so on, which needs to satisfy functional and aesthetic requirements. However, there is a relative lack of research on drone-built deep-seated systems, and the customer experience needs to be improved.

To increase user satisfaction with emergency rescue drone products, a product modelling design method based on the fuzzy Kano-QFD-FBS model is proposed. First, the initial user requirements of the emergency rescue drone products are obtained through a questionnaire, and the fuzzy Kano model is utilised and combined with the better–worse coefficient method to categorise the attributes, define the priorities of the user requirements, and screen out the key user requirements. Second, the QFD model is used to construct the quality house, analyse the key user requirements quantitatively, and obtain the design elements and weights of the emergency rescue drone product. The obtained key design elements are subsequently imported into the FBS model to complete the mapping transformation from the functional elements to the structural elements of the emergency rescue drone products and realise the styling design of the emergency rescue drone products. Finally, the user satisfaction scale based on appearance, functionality, and interaction was developed and the System Usability Scale (SUS) was used to evaluate user satisfaction with the emergency rescue drone design scheme. The new design scheme scored higher and showed significant differences in satisfaction ratings compared to the previous scheme. This study shows that the method of product styling design based on the fuzzy Kano-QFD-FBS model can comprehensively acquire and analyse user requirements, realise accurate mapping from user requirements to product design elements, and output the specific solution of the emergency rescue drone product styling design. The design scheme performs well in meeting user requirements, verifies the feasibility and effectiveness of the fuzzy Kano-QFD-FBS model in the styling design study of emergency rescue drones, and provides a new paradigm for emergency rescue product design. The framework of the basic content of the paper is shown in Figure 1.

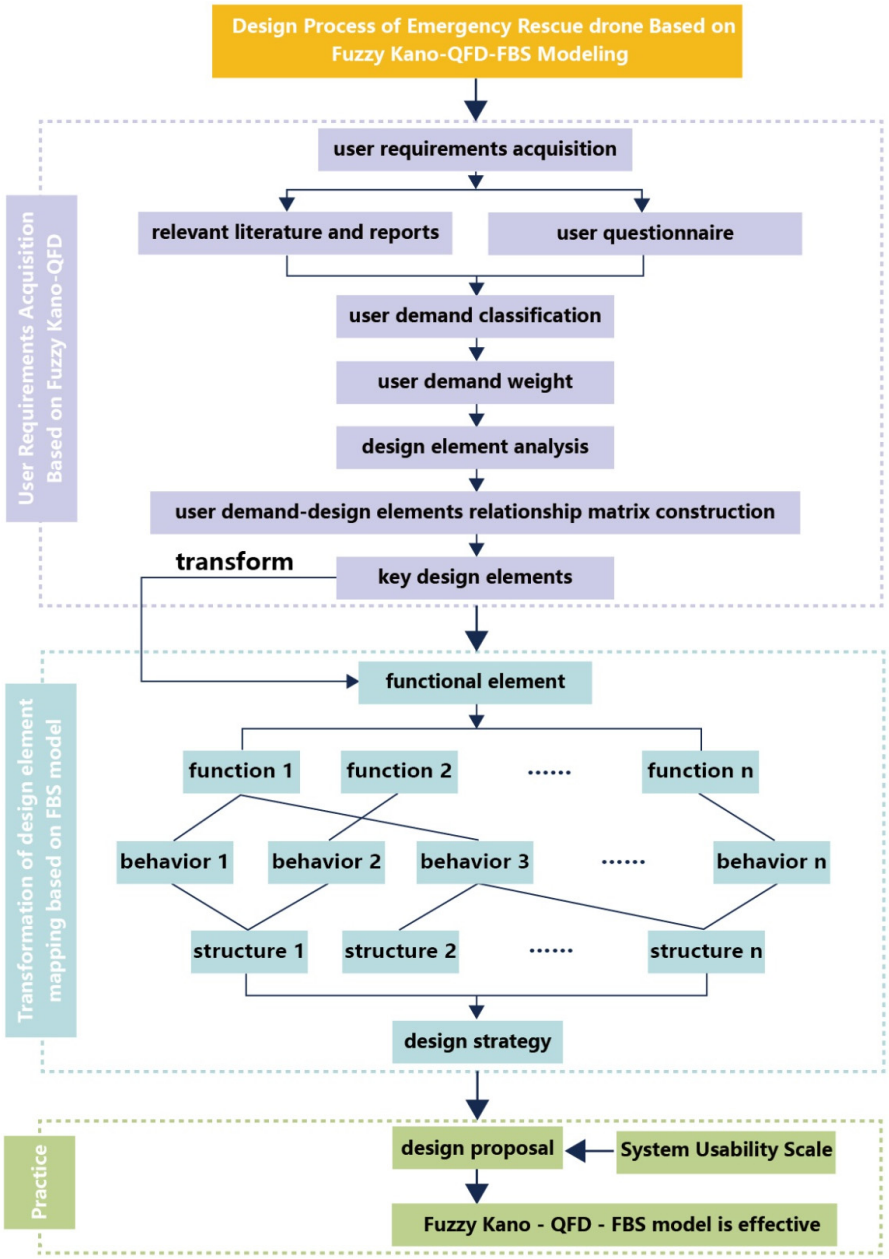


Figure 1. Framework of the basic content of the paper.

2. Literature Review

With the rapid development of online communication and image processing technology, the field of emergency rescue drones is expanding rapidly. Its mapping, monitoring, search, and other functions are increasingly applied in agriculture, mining, service experience, disaster relief, and other fields [15,16]. Especially in the case of danger and disaster relief, the emergency rescue drone can make preparations for the realisation of the emergency rescue drone in an unknown environment to compensate for the lack of access to disaster information in the first place [7]. Therefore, research on emergency rescue drones is highly important. Previous research on rescue equipment has focused mainly on technical systems, hardware integration, structural science, and research methods. Although

these studies have analysed different types and degrees of research, there is a lack of research on machine modelling, so a product modelling design method based on the fuzzy Kano-QFD-FBS model is proposed in this paper.

2.1. Fuzzy Kano Model

The Kano model is a type of household demand analysis method that is used to study the relationship between product characteristics and customer satisfaction [17–19]. Professor Noriaki Kano proposed the Kano model in 1984, which subdivides the quality elements into five categories [20], as shown in Figure 2.

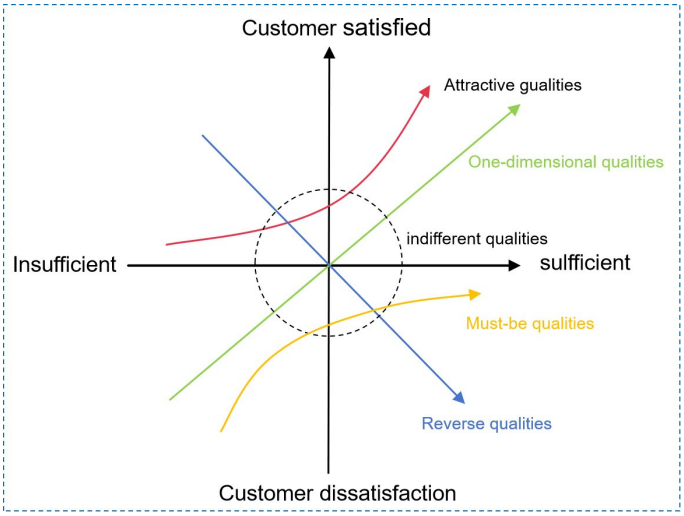


Figure 2. Kano model framework.

The model emphasises the importance of taking the household as the centre and understanding customer preferences [21]. When the Kano model is in response, the attributive relationship between the qualifying values 0 and 1 (0 means no, 1 means yes) makes accurate feedback on the actual needs of the customer difficult. There is usually a sense of certainty in the process of customer experience, so the introduction of fuzzy theory to the Kano model is an effective solution to the actual needs of households [22]. For example, when a certain quality element is available, according to the degree of customer preference, 1 is divided into 0.7, 0.2, and 0.1 (a sum equal to 1 is effective). The fuzzy Kano model questionnaire form is shown in Table 1.

Table 1. Fuzzy Kano model questionnaire.

Functions of the Product	Satisfied	Deserved	Indifferent	Reluctantly Accepted	Very Dissatisfied
Available for a particular function	0.7	0.2	0.1	-	-
Not available for a particular function	-	-	-	0.6	0.4

The fuzzy Kano model can effectively classify requirements into categories and rank their importance to ensure the rationality of the design results [23]. Zhang et al. [24] analysed review data via a product performance dictionary via the fuzzy Kano model and quickly identified customer requirements. Hao et al. [25] adopted the fuzzy Kano model to analyse the emotional design requirements of floor sweepers in the fault expression layer and improved the design of boundary vision and operation flow. Wang [26] used the fuzzy Kano model to divide household demand attributes and integrated emotional

elements into an intelligent air purifier design case. According to the data obtained from the fuzzy Kano model questionnaire, user requirements for functional attributes can be classified and summarised and combined with the better–worse analysis method [27]. The degree of household satisfaction with different functions can be evaluated, and their influence can be strengthened. Among them, the better coefficient (the increased satisfaction coefficient) is usually positive, which reflects the improvement in customer satisfaction after the realisation of a certain function of the product. Larger values indicate that the rate of user satisfaction for the feature increases faster with respect to a stronger positive user response to the feature. The worse coefficient (the dissatisfaction coefficient after elimination) is usually negative, indicating the rate at which customer satisfaction decreases when a feature of the product is removed or no longer meets customer demand. The larger the absolute value of the worst coefficient is, the greater the level of dissatisfaction with the missing feature, meaning that the absence of the feature would lead to a faster decline in user satisfaction. The better coefficient = (expectation + charm)/(basic + expectation + charm + charm difference), and the worse coefficient = (basic + expectation)/(basic + expectation + charm + charm difference).

2.2. QFD Model

QFD (quality function deployment), proposed by Professor Yoji Akao in 1972 [28], is a modern quality system. Its verification takes the customer as the starting point and can accurately transform the subjective needs of the customer into the design requirements of the product [29]. Thus, products can meet the cognitive needs of customers, which has always been the focus of product design research. Lv et al. [30] enabled the design of a basic requirement model of the QFD and improved the efficiency of draft selection and aggregation. Yu and Wang [31] transformed the cognitive needs of residents into quality improvement elements of health examination services by using QFD methods and improved the service management quality of hospitals. Xu et al. [32] used the QFD and FBS models to study the multi-suitability design of toilets so that the objective performance of the final product and the cognitive needs of the customer can be accurately mapped and matched. Zhou et al. [33] used an integrated model of QFD and an orthogonal design of DOE experiments to improve the service quality and user satisfaction of medical sterilisation products by transforming the obtained customer requirements into engineering characteristics to provide decisions for product design and development. QFD is a quality management methodology that combines qualitative and quantitative analyses. Driven by user requirements, the QFD focuses on quality management at the product design stage to ensure product quality at the source. The QFD model structure is shown in Figure 3.

The QFD method based on the fuzzy Kano model first classifies the user requirements through the fuzzy Kano model, calculates the satisfactory and unsatisfactory aspects of each type of need, and takes the weighted or maximum value of the two for the percentage of needs satisfied, which is normalised to be the customer demand weight of the QFD [34,35].

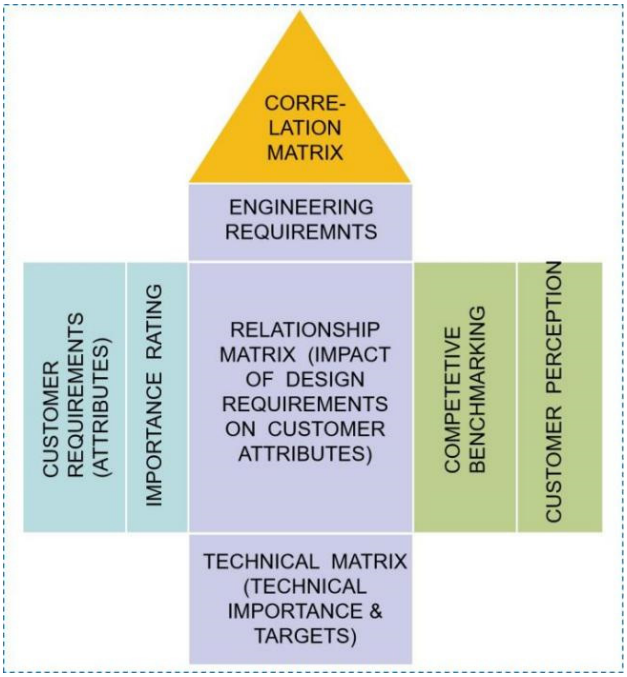


Figure 3. QFD model framework.

2.3. FBS Model

The FBS model is a concept proposed by Gero et al. The model focuses on mapping transformation through function–behaviour–structure and assists designers in the development of innovative product designs [36,37]. In this model, functions, behaviour, and structures are embodied in three "worlds": the objective world, the interpreted world, and the expected world. The initial 8 steps connecting functions, behaviour, and structures are reconstructed to produce a new set of 20 steps that together form the Scenario FBS (SFBS) model shown in Figure 4.

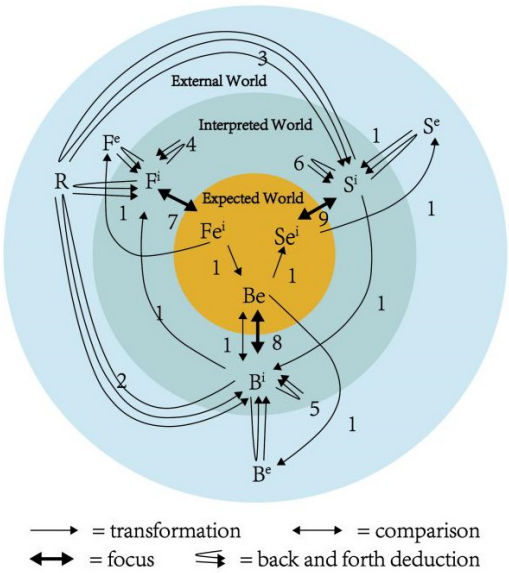


Figure 4. The situated FBS model framework.

The FBS model focuses on mapping transformation through function–behaviour–structure and assists designers in innovative product design development. The FBS model

realises the mapping process from product function (F) to domain (B) through the analysis of operation and then maps to specific design requirements (S) to improve the reasonableness of the product conceptual design stage. Wang et al. [38] built an innovative furniture design process based on situation analysis and the FBS model to solve the problem of mismatch between the design scheme and the actual needs of users over a long period of innovative furniture design. Zhou et al. [39] proposed an integrated innovative design method based on fuzzy Kano and scene FBS models and used the scene of saving a baby elephant toy as an example to verify the feasibility of the innovative design method. Chen et al. [40] proposed an innovative design mode that integrates the QFD and FBS to realise the innovative design of mobile power detection equipment.

The FBS model comprises three variables: function, behaviour, and structure, with hierarchical mapping among the three variables. The FBS model is employed to realise the mapping process from product function to behaviour and mapping to specific design requirements through the analysis of human operation behaviour to improve the rationality of the product conceptual design stage. The FBS model is an important tool for design innovation, as it generates a number of conceptual formulations for the target problem, providing designers with ideas on the basis of the needs of the subject and the final function and structure of the solution.

2.4. Fuzzy Kano-QFD-FBS Model Fusion Application

At present, research on the fusion application of the Kano, QFD, and FBS models has achieved certain results, which have been applied to product design, interaction design, and service experience design. Büyüközkan et al. [41] developed an integrated method that consists of the QFD, Kano model, and interval-valued intuitionistic fuzzy analytical hierarchy to provide a customer-oriented product design. Wu et al. [42] developed a function that combines the baby stroller design method with the Kano, QFD, and FAST methodologies. Rampal et al. [43] used Kano and QFD analyses for autonomous electric cars to increase customer contentment. Chen et al. [44] proposed an online review-driven Kano-QFD method for service design. Li et al. [45] used dissected interviews and questionnaire surveys to introduce the Kano-QFD research method into UX design research in the area of handicraft ICH apps. Zhao and Chen [46] used TRIZ theory with the Kano model and QFD theory to design helmet products. Tandiono and Rau [47] proposed an enhanced model using the Kano model, QFD, and TRIZ with a component-based approach for systematically designing sustainable and innovative products. Wan et al. [48] used an innovative engineering design process integrated with the QFD and FBS model to assist in the design of transport devices. Wang et al. [49] researched the design of children's swimming auxiliary products via the QFD and FBS models, enabling precise mapping and matching between the objective performance of the final product and user requirements. Li et al. [50] combined the QFD theory with the FBS model and applied it to the preliminary design of a braille writing modifier, transforming subjective abstract user requirements into concrete structured information and verifying the feasibility of the integrated application of the two in accessible design.

The above research focuses mainly on the integration of the Kano-QFD model or the QFD-FBS model, while there is less research on the integration of the fuzzy Kano-QFD-FBS model. The fuzzy Kano model can screen out key user requirements, ensuring that the design process closely aligns with user requirements. The QFD model can construct a quality house, conduct quantitative analysis on key user requirements, and obtain product design elements and weights. The FBS model can accurately analyse key design elements and translate them into objective and specific design guidance, solving the problems of high subjectivity and low product matching in the product design process. The

integration of the fuzzy Kano-QFD-FBS model can comprehensively capture user requirements, achieve accurate mapping from user requirements to product design elements, and translate into objective and specific design guidance. Therefore, this article combines the fuzzy Kano-QFD-FBS model to carry out research on the styling design of emergency rescue drones.

3. Method

To increase user satisfaction with emergency rescue drone products, a product modelling design method based on the fuzzy Kano-QFD-FBS model is proposed.

3.1. Fuzzy Kano User Requirements

Step 1: Fuzzy Kano questionnaire design. The product was investigated and analysed through a literature query, a questionnaire survey, and a user interview, and an evaluation item table of product user requirements was established, as shown in Table 2.

Table 2. Evaluation item list of user requirements.

Primary Indicators	Secondary Indicator
F_1	F_{11}
F_1 represents the first primary indicator.	F_{12}
	F_{1n}
.	.
.	.
.	.
F_n	F_{n1}
F_N represents the Nth primary indicator.	F_{n2}
	F_{m}

Each user requirement is scored with respect to both positive and negative aspects, and each user requirement feeling is divided into five levels: "satisfied", "deserved", "indifferent", "reluctantly accepted", and "very dissatisfied". The appropriate parameters are selected according to the degree of satisfaction (0–1), and the total score of the forward question and the reverse question of each user's requirements is 1, as shown in Table 1.

Step 2: Fuzzy Kano data analysis. The fuzzy Kano questionnaire with two positive and negative questions is designed according to the selected user requirements evaluation items, and the interactive evaluation matrix Q is established. By combining the interactive evaluation matrix Q with the fuzzy Kano model evaluation table (Table 3), the membership values T_{F11} [51] of the user requirements items are calculated. In the Table 3, *M, O, A, Q,* R and I, respectively, represent basic demand, expectation demand, charm demand, suspicious demand, reverse demand, and undifferentiated demand. Basic demand, expectation demand, charm demand, and undifferentiated demand are the effective results that need to be retained. In contrast, inverse and suspicious requirements need to be excluded in the statistical process. Suspicious demand means that users are satisfied or very dissatisfied with both sides of the same question, showing a contradictory attitude. In general, if the survey questions are properly designed and understood by the user, the data collected will be reliable. A suspicious result can mean a flaw in the question design, a mistake in the user's understanding of the question, or a mistake in the user's answer.

Table 3. Fuzzy Kano user requirements categorisation table.

Functions of the Product		Not Available for a Particular Function				
		Satisfied	**Deserved**	**Indifferent**	**Reluctantly Accepted**	**Very Dissatisfied**
Available for a particular function	Satisfied	*Q*	*A*	*A*	*A*	*O*
	Deserved	*R*	*I*	*I*	*I*	*M*
	Indifferent	*R*	*I*	*I*	*I*	*M*
	Reluctantly accepted	*R*	*I*	*I*	*I*	*M*
	Very dissatisfied	*R*	*R*	*R*	*R*	*Q*

Step 3: The fuzzy Kano model is quantified to obtain the importance of the end-user requirements. First, the user satisfaction coefficient is calculated, and the importance of the user requirements items is ordered. The user satisfaction coefficient [27] includes a satisfaction index (S_i) and a dissatisfaction index (D_i). Equation (1) is used to obtain S_i and D_i.

$$S_i = \frac{A_i + O_i}{A_i + O_i + M_i + I_i}, D_i = (-1)\frac{M_i + O_i}{A_i + O_i + M_i + I_i} \quad (1)$$

S_i and D_i represent the positive impact of a certain demand element on improving user satisfaction and eliminating the negative impact of user dissatisfaction, respectively. The values of these variables range from 0. When S_i is close to 1, the demand factor has a significant effect on improving user satisfaction. When S_i is close to 0, the effect is worse. Similarly, when D_i is close to −1, the demand factor has a very significant effect on eliminating user dissatisfaction. When D_i is close to 0, the effect is worse. The impact of user requirements can be measured by the maximum satisfaction index (T_i). Equation (2) is used to obtain the maximum satisfaction index.

$$T_i = \max\left(|S_i|, |D_i|\right) \quad (2)$$

To accurately measure the importance of user requirements, the importance index *K* is introduced. Depending on the type of requirement, different *K* values are assigned to different types of requirements. According to the research results of Chaudha A et al. [52], the *K* values of necessity demand, expectation demand, and charm demand are 0.5, 1, and 1.5, respectively. This grading ensures that we accurately quantify the priority of the user requirements.

Second, the user requirements adjustment improvement rate (IR_0) is calculated. To evaluate market competitiveness, a Likert scale was used to calculate users' current satisfaction with different demand items (S_0) and target satisfaction (S_i). By comparing the gap between the target satisfaction and the current satisfaction, the target improvement rate of user satisfaction (IR_0) is calculated. Equation (3) is used to obtain IR_0.

$$IR_0 = S_i / S_0 \quad (3)$$

The demand adjustment improvement rate (IR_i) is calculated on the basis of the user satisfaction index (T_i) and the target improvement rate (IR_0). Equation (4) is used to obtain IR_i.

$$IR_i = (1 + T_i)^K \cdot IR_0 \quad (4)$$

Finally, the importance of the end-user requirements (W_i) is calculated. By adjusting the improvement rate (IR_i) and self-scoring the importance of the user requirements according to the needs (H_i), we obtain the importance of the end-user requirements (W_i) after the improvement. Equation (5) is used to obtain W_i.

$$W_i = IR_i \cdot H_i \tag{5}$$

3.2. Establishing the QFD Quality Housing Model

The above user requirements are transformed, the design elements are output, the corresponding design requirement targets are summarised and integrated, and a QFD quality house model for the product is built. QFD quality house matrix modelling is performed. As shown in Figure 5. The "left wall" of the quality house represents the user's needs, and the "ceiling" represents the design elements. The correlation between the user requirements and design requirements represented by each group is represented by a different score. The degree of correlation between the user requirements (F_i) and design elements (E_i) is R_{ij}. The design requirements importance (n_j) and relative importance (v_j) are calculated via Equation (6).

$$n_j = \sum_{i=1}^{n} w_i \cdot R_{ij}, v_i = n_j / \sum_{j=1}^{m} n_j \tag{6}$$

where i = 1,2,…,n, n represents the number of user requirements, j = 1,2,…,m, and m represents the number of design elements. By scoring the strength of the correlation between user requirements and design elements, we establish the relationship between them. A strong correlation is represented by ●, with the highest score being 5 points. A medium correlation is represented by ○, with a median score of 3 points. A weak correlation is represented by Δ, with the lowest score being 1 point. No correlation is represented by a space.

	importance degree	E_1	E_2	...	...	...	E_m
F_1							
F_2							
...							
...							
...							
F_n							
	n_j	n_1	n_2	...	...	...	n_m
	v_j	v_1	v_2	...	...	...	v_m
	srot						

Figure 5. Relationship matrix between user requirements and design elements.

According to Equation (6), the weight of each element is calculated, and the key points in the design process are determined successively to avoid the design direction deviating from the user requirements and design elements. In the design planning stage, priority should be given to ranking the design elements according to their importance and analysing them one by one. We need to be clear about the interconnections and influences between these elements and their specific guidance on design direction to ensure that the final design truly meets the needs of the user.

3.3. Mapping Transformation of Design Elements via the FBS Model

After the total function of the product is obtained via the fuzzy Kano model, quantitative analysis is carried out via quality function expansion, and the requirement is transformed into a design factor.

Step 1: Functional element transformation. According to the above calculation, the design elements with higher weights are summarised as key design elements, converted into functional elements, and imported into the FBS model. Material- and appearance-related design elements are excluded from the FBS mapping transformation.

Step 2: Function–behaviour–structure mapping transformation. Function–behaviour mapping transforms the obtained functional element mapping into the user's specific expected behaviour. Behaviour–structure mapping translates the expected behaviour mapping into the concrete structure of the product. The contradictions between various behaviour should impose certain limitations on these structures to avoid conflict. A behaviour may be completed by multiple parts, so it is necessary to consider the scale of the user's behaviour and arrange these structures properly. A structure can also have multiple functions. It is necessary to consider the similarities and differences between the two behaviour and generate innovative structures via analogical reasoning methods.

Step 3: Design strategy analysis on the basis of FBS mapping transformation. On the basis of the results of the FBS mapping transformation, objective and specific design strategies are proposed to guide designers in modelling design.

4. Empirical Case

Since the beginning of the 21st century, drones have been widely used in various fields, and their shapes have become increasingly diverse to meet the dual needs of function and aesthetics. To fully explore and transform user requirements and clarify the direction of emergency rescue drone modelling design, the fuzzy Kano-QFD model is adopted to obtain macro design elements to ensure that the design process closely conforms to the actual requirements of users. The FBS model can provide accurate design analysis and transform it into objective and specific design guidance. The problems of high subjectivity and low product matching in the design process of emergency rescue drones should be solved. The comprehensive application of the model based on the fuzzy Kano-QFD-FBS can provide a new scientific way to model the design of emergency rescue drones.

4.1. Fuzzy Kano User Requirements Analysis

Step 1: Fuzzy Kano questionnaire design. First, through a literature search, questionnaire survey, and user interviews, the emergency rescue drone products were investigated, and 36 user requirements for emergency rescue drones were obtained. Then, 19 user requirement evaluation items were selected via the expert opinion method and divided into three levels: modelling design, functional design, and interaction design. The user requirements are numbered in the form of F_1–F_3, as shown in Table 4.

Table 4. User requirements assessment projects.

Primary User Requirements	Secondary User Requirements Serial Number	Meaning of Secondary User Requirements Serial Number
Styling design F_1	F_{11}	Simple styling
	F_{12}	Proportional coordination
	F_{13}	Striking colours

	F_{14}	Appropriate material
	F_{15}	Fine workmanship
Functional design F_2	F_{21}	Auxiliary fire extinguishing
	F_{22}	Material input
	F_{23}	Lighting functions
	F_{24}	Broadcast guidelines
	F_{25}	Payload carrying
	F_{26}	Longer battery life
	F_{27}	Information collection
	F_{28}	Real time monitoring
	F_{29}	Autonomous navigation
Interaction design F_3	F_{31}	Convenient operation
	F_{32}	Status indicator
	F_{33}	Real-time feedback
	F_{34}	Intelligent decision-making
	F_{35}	One-click return

Step 2: Fuzzy Kano data analysis. The fuzzy Kano questionnaire with two positive and negative questions is designed according to the selected user requirements evaluation items. A total of 100 questionnaires were sent out, and 89 valid questionnaires were recovered, which is highly reliable. User feedback data are obtained through questionnaire analysis. Taking the concise shape of F_{11} as an example, a fuzzy questionnaire was established, as shown in Table 5.

Table 5. Fuzzy questionnaire.

Simple Styling	Satisfied	Deserved	Indifferent	Reluctantly Accepted	Very Dissatisfied
The product has this element	0.7	0.2	0.1	0	0
The product does not have this element.	0	0	0.1	0.3	0.6

When the product has this element, the matrix of establishment is X = [0.7 0.2 0.1 0 0]. When the product does not have this element, the matrix is Y = [0 0 0.1 0.3 0.6], and the fuzzy interaction matrix is as follows:

$$Q = X^T Y = \begin{bmatrix} 0 & 0 & 0.07 & 0.21 & 0.42 \\ 0 & 0 & 0.02 & 0.06 & 0.12 \\ 0 & 0 & 0.01 & 0.03 & 0.06 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (7)$$

The distribution of the median value of the matrix Q_{F11} corresponds to Table 5, and the class membership vector T_{F11} of the user requirements attribute with a concise shape is obtained as follows:

$$T_{F11} = \left[\frac{0.18}{M}, \frac{0.42}{O}, \frac{0.12}{I}, \frac{0.28}{A}, \frac{0}{R}\right] \quad (8)$$

where Q_M represents the sum of the values of the *M* attributes in the matrix, which is the membership vector of Q_M. $t_M = 0.18$. By calculating separately for the other four types of requirements, Q_O、Q_I、Q_A, and Q_R, the Q_{F11} membership vectors are $t_M = 0.18, t_O = 0.42, t_I = 0.12, t_A = 0.28$, and $t_R = 0$, respectively.

In order to obtain more reliable data, threshold α is introduced to screen the data in the vector obtained from the questionnaire results. If the value of a factor is ≥α, the factor is represented by 1, no 0. Here, we set the threshold to 0.42. T_{F1} is calculated at this time.

$$T_{F11} = [0,1,0,0,0] \quad (9)$$

To analyse the attributes of each requirement element, the following steps are taken:

Step 1: For each demand element, the cumulative number of its attributes is calculated first.

Step 2: The demand element is defined according to the attribute with the highest cumulative number.

Step 3: If it is found in the final statistics that a certain demand element has the same cumulative number of multiple attributes, it is selected according to the priority order of attributes (*M, O, A, I, R* from high to low).

Step 4: In questionnaire statistics, if there are multiple attributes in the same demand factor, the membership vector t of these attributes is compared, and the attribute with the larger value is selected.

Step 5: If the membership vectors T of the attributes are equal, the attributes are characterised according to their priority order.

The statistical results of the user requirements are shown in Table 6. According to the statistical results, F_{11}–F_{13} belong to the attractive demand category. F_{14}, F_{25}, F_{27}–F_{29}, and F_{33}–F_{34} are basic requirements. F_{15}, F_{21}–F_{24}, F_{26}, F_{31}, and F_{35} belong to the charm requirements. F_{32} is an undifferentiated requirement. According to the classification results of the fuzzy Kano model, the essential requirements are basic requirements, and the additional functional requirements are mostly charm requirements and expectation requirements. Since the undifferentiated requirements of F_{32} have no influence on the satisfaction of the design scheme, this design is not considered.

Table 6. Statistical categories.

Primary User Requirements	Secondary User Requirements Serial Number	Design Element	Form
Styling design F_1	F_{11}	Simple styling	*A*
	F_{12}	Proportional coordination	*A*
	F_{13}	Striking colours	*A*
	F_{14}	Appropriate material	*M*
	F_{15}	Fine workmanship	*O*
Functional design F_2	F_{21}	Auxiliary fire extinguishing	*O*
	F_{22}	Material input	*O*
	F_{23}	Lighting functions	*O*
	F_{24}	Broadcast guidelines	*O*

	F_{25}	Payload carrying	*M*
	F_{26}	Longer battery life	*O*
	F_{27}	Information collection	*M*
	F_{28}	Real time monitoring	*M*
	F_{29}	Autonomous navigation	*M*
Interaction design F_3	F_{31}	Convenient operation	*O*
	F_{32}	Status indicator	*I*
	F_{33}	Real-time feedback	*M*
	F_{34}	Intelligent decision-making	*M*
	F_{35}	One-click return	*O*

Step 3: The fuzzy Kano model is quantified to obtain the importance of the end-user requirements.

The user satisfaction coefficient is calculated. The importance of the user requirements items is ranked and calculated via Equations (1) and (2) to obtain the value of each demand element, as shown in Table 7.

Table 7. User demand satisfaction index and adjustment factor.

Secondary User Requirements Serial Number	S_i	D_i	T_i	K
F_{11}	0.70	−0.60	0.70	1.5
F_{12}	0.62	−0.43	0.62	1.5
F_{13}	0.59	−0.42	0.59	1.5
F_{14}	0.48	−0.83	0.83	0.5
F_{15}	0.57	−0.53	0.57	1
F_{21}	0.60	−0.70	0.70	1
F_{22}	0.60	−0.51	0.60	1
F_{23}	0.65	−0.70	0.70	1
F_{24}	0.69	−0.66	0.69	1
F_{25}	0.40	−0.50	0.50	0.5
F_{26}	0.60	−0.53	0.60	1
F_{27}	0.39	−0.59	0.59	0.5
F_{28}	0.44	−0.65	0.65	0.5
F_{29}	0.52	−0.65	0.65	0.5
F_{31}	0.60	−0.65	0.65	1
F_{33}	0.45	−0.65	0.65	0.5

F_{34}	0.55	−0.69	0.69	0.5
F_{35}	0.63	−0.79	0.79	1

The target improvement rate of the user demand is calculated, and the improvement rate and importance are adjusted via the following equation: Equations (3)–(5) were used to calculate the target improvement rate of the user requirements and adjust the improvement rate and importance. Table 8 shows the values of each indicator of the requirements.

Table 8. Target improvement rate, adjusted improvement rate, significance.

Secondary User Requirements Serial Number	Target Improvement Rate IR_o	Adjusted Improvement Rate IR_i	Significance W_i
F_{11}	1.41	3.12	11.40
F_{12}	1.46	2.06	8.77
F_{13}	1.47	2.94	10.88
F_{14}	1.59	2.15	9.59
F_{15}	1.50	2.36	9.96
F_{21}	1.56	2.65	12.24
F_{22}	1.49	2.38	10.52
F_{23}	1.56	2.65	10.97
F_{24}	1.55	2.62	11.11
F_{25}	1.44	1.76	6.48
F_{26}	1.49	2.38	10.95
F_{27}	1.49	1.88	7.60
F_{28}	1.52	1.95	8.74
F_{29}	1.53	1.97	8.63
F_{31}	1.54	2.54	11.84
F_{33}	1.53	1.97	8.47
F_{34}	1.54	2.00	8.44
F_{35}	1.58	2.83	12.06

4.2. Establishing the QFD Quality House Model

The above user requirements are transformed, the design elements are output, and the corresponding design requirements and objectives are summarised and integrated, as shown in Table 9.

Table 9. Design element objectives.

Primary Design Element	Secondary Design Element Serial Number	Meaning of Secondary Design Element	Design Element Objectives
Appearance and structure E_1	E_{11}	Morphological design	Simple and beautiful modelling
	E_{12}	Colour scheme	Simple colour scheme with striking colour
	E_{13}	Material selection	The material is resistant to high temperature, corrosion and waterproof
Functional performance E_2	E_{21}	Auxiliary fire extinguishing	Carrying equipment such as fire-fighting rounds
	E_{22}	Material input	Delivery of items such as masks, medication, etc.
	E_{23}	Lighting functions	Low visibility, night lighting
	E_{24}	Broadcast guidelines	Equipped with voice guidance
	E_{25}	Payload carrying	Webcam and more
	E_{26}	Longer battery life	High-capacity battery
Intelligent features E_3	E_{31}	Information collection	Flight status, ground test measurements
	E_{32}	Real time monitoring	Aerial photography management, environmental monitoring data analysis
	E_{33}	Autonomous navigation	Route planning, heading control and mission execution
	E_{34}	Real-time feedback	Instantaneous transmission of flight data to the control system
	E_{35}	Intelligent decision-making	Strong skills in task planning, execution, etc.
User experience E_4	E_{41}	Convenient operation	Simplified control interface
	E_{42}	Status indicator	Indicator colours and blinking patterns convey information
	E_{43}	One-click return	Safe return to take-off point in case of emergency

According to Equation (6), the relationship matrix between the user requirements and design elements is established, as shown in Figure 6.

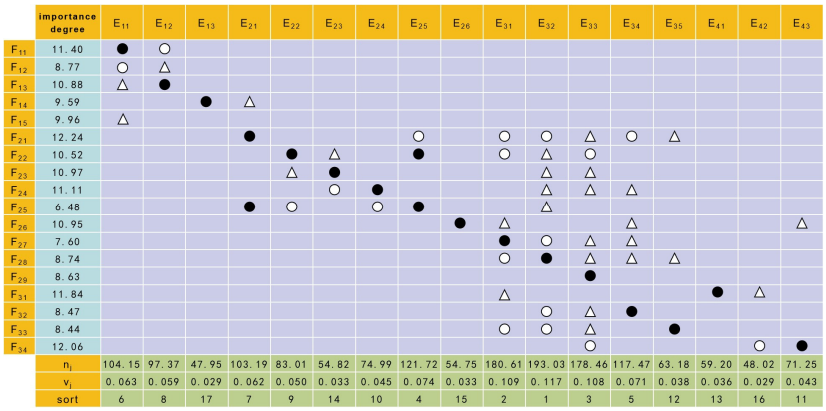


Figure 6. Relationship matrix of emergency rescue Drones

An analysis of Figure 5 reveals that the importance value of E_{32} real-time monitoring data is 0.117, which ranks first. E_{11} has different degrees of correlation with F_{21} auxiliary fire suppression, F_{22} material delivery, F_{23} lighting function, F_{24} broadcast guidance, F_{25} load carrying, F_{27} information collection, F_{28} real-time monitoring, F_{17} real-time feedback, and F_{34} intelligent decision-making in the quality house. Therefore, the design should be integrated with high-definition cameras and supplemented with lighting functions to ensure that clear images can be provided under a variety of light conditions.

The importance value of the ground measurement of the E_{31} information acquisition flight status is 0.109, which ranks second. E_{31} has different degrees of correlation with F_{21} auxiliary fire extinguishing, F_{22} material delivery, F_{26} long endurance, F_{27} information collection, F_{29} autonomous navigation, F_{31} convenient operation, and F_{34} intelligent decision-making in the quality house. Therefore, in the design, it is necessary to combine the working environment of the drone field to ensure the normal operation of the drone system and the accuracy of data acquisition.

The importance value of the E_{33} autonomous navigation planning path is 0.108, which ranks third. E_{33} has different degrees of correlation with F_{21} auxiliary fire suppression, F_{22} material delivery, F_{23} lighting function, F_{24} broadcast guidance, F_{27} information collection, F_{28} real-time monitoring, F_{33} real-time feedback, F_{18} intelligent decision-making, and F_{35} one-click return in the quality house. Therefore, on-board sensors and computing power are used in the design to ensure self-positioning, path planning, and obstacle avoidance.

4.3. Mapping Transformation of Design Elements

After the total function of the product is obtained via the fuzzy Kano model, quantitative analysis is carried out via quality function expansion, and the requirements are transformed into a design factor.

Step 1: Function–behaviour mapping. On the basis of the FBS model, various performances of the emergency rescue drone are analysed, and various functions, such as communication, flight, reconnaissance, and assisted rescue, are obtained. The functions are refined and become corresponding behaviour through mapping. The communication function is mapped as voice interaction and real-time video transmission. The flight function is mapped to automatic route planning and low battery return. The reconnaissance function is mapped to fire monitoring, toxic gas detection, and smoke concentration detection. The rescue function is mapped to an autonomous drone rescue system and image transmission. To ensure the safety of trapped people, lighting and broadcasting guidance modules are added to meet the demand for safe escape with low visibility.

Step 2: Behaviour–structure mapping. In the behaviour mapping structure, each actor-mapped module is related to another in such a way that the actors and the structure

are likely to be mapped in a one-to-many relationship. The product structure layout is more reasonable and easier to master. The corresponding structure is mapped according to the behaviour of the emergency rescue drone. The mapping result of the function–behaviour structure of emergency rescue drones is shown in Figure 7.

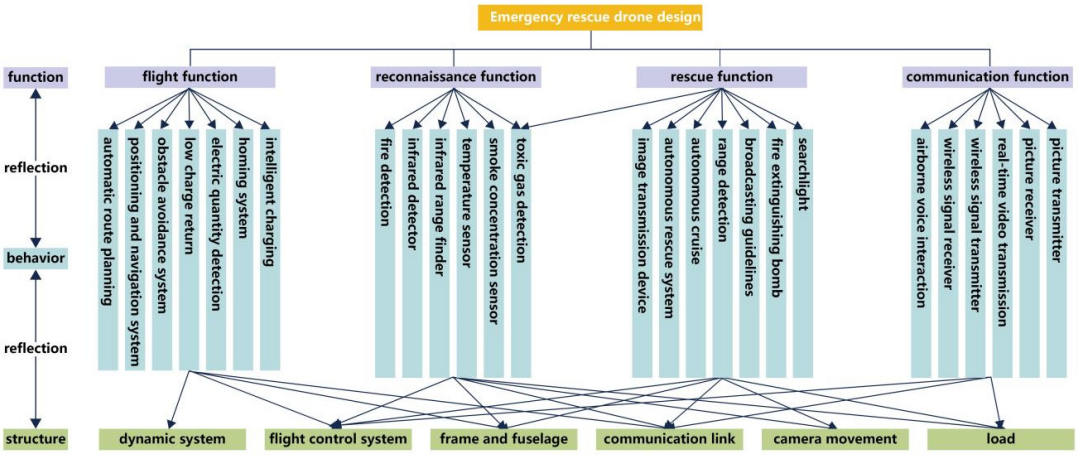


Figure 7. Function–behaviour–structure mapping.

4.4. Design Scheme

The modelling software Rhino 8 was used to complete the 3D model production, and the Keyshot11 rendering software was used to complete the rendering of the model. The model was made more realistic by utilising the model material, colour, texture, lighting, etc. The visualisation of the model can be highlighted by adjusting the parameters. The colour with high recognition is used according to the characteristics of the product workplace, which is mainly orange with other colours. The renderings are shown in Figure 8.



Figure 8. Emergency rescue drone renderings from different angles.

5. Results Verification and Analysis

Hefei Jiaxun Technology Co., Ltd. carried out product development for the design scheme. The emergency rescue drone's detailed specifications are as follows. Weight: 6470 g (with dual batteries). Longest flight time: 55 min. Environmental perception: front, back, left, right, up and down, visual + infrared sensors, six-way obstacle avoidance. Environmental adaptability: IP55 protection, maximum take-off altitude of 7000 m, operating temperature range of −20 °C to 50 °C. Maximum wind resistance speed: 12 m/s. Payload capacity: maximum payload of 2.7 kg, fully meeting different operational needs in public safety, inspection, mapping, etc. In terms of intelligent decision-making, the drone can autonomously judge the fire situation and make firefighting decisions. In addition, there are artificial intelligence-based fire rescue command and decision-making methods, including receiving the original fire scene images captured by the firefighting drone and processing the fire scene images to obtain the fire scene image. In terms of autonomous navigation, the drone can achieve autonomous navigation in tunnel environments without satellite signals, accurately locating the fire source and personnel, and quickly deploying rescue equipment or firefighting materials. By using GPS positioning and intelligent navigation technology, it can accurately fly to the fire point.

At present, physical products have been sold on the market and have achieved good results. To verify the actual degree of satisfaction of the design scheme for user requirements, a user satisfaction rating scale was established from three dimensions: functionality, appearance and interaction. The system usability scale (SUS) [53] was adopted to compare the user experience desirability test results between the products designed in this paper and those previously designed by the company. The specific test contents are shown in Tables 10 and 11. The user satisfaction rating scale in the above two rating scales is established based on expert review and evaluation

Table 10. User satisfaction rating scale.

Number	Dimension	Subject	Strongly Dissatisfied	Dissatisfied	Usual	Satisfied	Strongly Satisfied
1	Appearance dimension	Satisfaction to the modelling of the product	0	1	2	3	4
2		Satisfaction to the colour of the product	0	1	2	3	4
3		Satisfaction to material quality	0	1	2	3	4
4	Function dimension	Satisfaction to the transportation function	0	1	2	3	4
5		Satisfaction to the battery life	0	1	2	3	4
6		Satisfaction to the data transmission function	0	1	2	3	4
7		Satisfaction to the path planning and navigation system	0	1	2	3	4
8	Interaction dimension	Satisfaction to operating system	0	1	2	3	4

9	Satisfaction to alarm feedback system	0	1	2	3	4

Two types of survey questionnaires were distributed to 200 users each, and 196 and 194 valid questionnaires were collected, respectively. Reliability and validity testing of the questionnaire was carried out, and the scoring criteria were as follows:

Table 11. SUS questionnaire.

Number	Subject	Strongly Disagree	Disagree	Usual	Agree	Strongly Agree
1	Regular use of emergency response drones	0	1	2	3	4
2	Worried about using it	0	1	2	3	4
3	It is easy to use	0	1	2	3	4
4	Need guidance or assistance in use	0	1	2	3	4
5	Pleasant to use	0	1	2	3	4
6	Learning is required to use	0	1	2	3	4
7	Emergency rescue drone features are well integrated	0	1	2	3	4
8	Emergency response drones do not have to be high-tech	0	1	2	3	4
9	Emergency rescue drones are beautifully styled	0	1	2	3	4
10	Emergency rescue drones are strangely shaped	0	1	2	3	4

Step 1: Determine the conversion score for each question, which ranges from 0 to 4. For positive questions (odd questions), the conversion value is the original score of the scale minus 1 ($X_i - 1$). For negative (even) questions, the conversion score is 5 minus the original score ($5 - X_i$).

Step 2: The conversion scores of all the items are added and multiplied by 2.5 to obtain the total score of the SUS scale, which ranges from 0 to 100.

Step 3: Based on the scoring results, the reliability and validity of the survey questionnaire are calculated. The SUS scale is a globally recognised usability evaluation scale, and its reliability and validity have been fully tested. The reliability of the satisfaction rating scale is determined based on the Cronbach's alpha coefficient, as shown in Table 12. After testing, the Cronbach's alpha coefficient of the satisfaction rating scale is 0.845, indicating reliability.

Table 12. Cronbach's reliability coefficient value scale.

Reliability Range	Reference Standards
0.9 ≤ Cronbach α	The internal reliability of the reference standard is very high and reliable
0.8 ≤ Cronbach α < 0.9	High internal reliability, trustworthy

0.7 ≤ Cronbach α < 0.8	Acceptable internal reliability, trustworthy
0.6 ≤ Cronbach α < 0.7	The scale has certain issues, but it has reference value and is reliable
Cronbach α < 0.6	The scale has significant issues and needs to be redesigned

Step 4: The structural validity of the questionnaire is verified through factor analysis, and the specific process is as follows: The KMO value and Barlett test statistic calculated in SPSS based on the scoring results are indicators used to compare the simple correlation coefficient and partial correlation coefficient between variables. After calculation, the KMO value and Bartlett value of the questionnaire were 0.841 (KMO > 0.8) and 0.03 (Sig < 0.05), respectively, indicating that the validity test of the questionnaire is suitable for factor analysis. Exploratory factor analysis (EFA) was used to establish the correspondence between items and factors. Through calculation, it was found that the factor values of 8 items were all greater than 0.4, indicating that they were basically consistent with previous expectations and the questionnaire validity was good.

Step 6: The original score value is converted into a percentile grade; referring to Table 13. The corresponding rating is found and the score of the test object is obtained.

Table 13. Range of curve grading for the total SUS score.

SUS Total Score Grade	Rating	Percentage Rating	SUS Total Score Grade	Rating	Percentage Rating
84.1–100	A+	96–100	71.5–72.5	C+	60–64
80.8–84.0	A	90–95	65–71	C	41–59
78.9–80.7	A−	85–89	62.7–64.9	C−	35–40
77.2–78.8	B+	80–84	51.7–62.6	D	15–34
74.1–77.1	B	70–79	0–51.7	F	0–14
72.6–74	B−	65–69			

Collect user satisfaction ratings and SUS scale ratings and establish statistical descriptions of the data results from several dimensions in Table 14.

Table 14. Description and statistics of user rating data.

Group	Sample Scale	Minimum	Maximum	Mean	Standard Deviation	Median
Current Product Questionnaire (A)	98	28	35	32.02	3.85	32
Original Product Questionnaire (B)	98	17	25	19.53	4.36	20
Current product SUS (C)	97	71	87	80.04	6.72	78
Original product SUS (D)	97	36	58	48.82	7.36	45

The results show that the emergency rescue drones designed in the paper has a high score on the SUS scale, reaching more than 80 points, as shown in Figure 9. This shows that the design scheme of the emergency rescue drone can effectively meet the actual needs of most users and provide them with pleasant experiences.

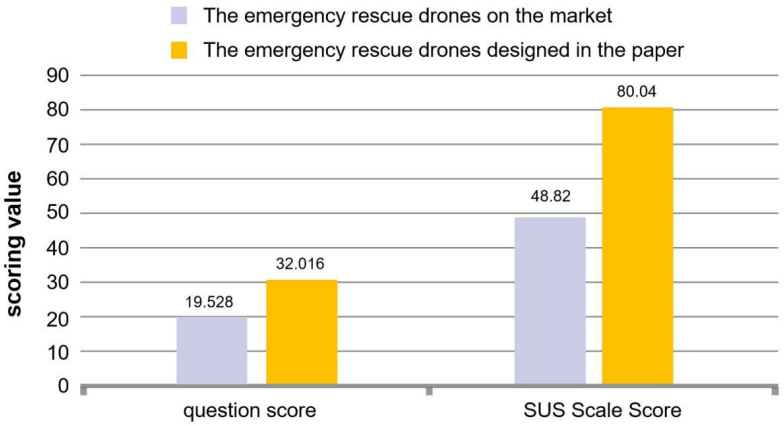


Figure 9. SUS test results.

Differential tests were conducted on groups AB and CD, respectively. As the data met the requirements of normal distribution test and homogeneity of variance, independent sample test was selected to calculate the difference between the two groups, with *p*-values of 0.01 and 0.03 ($p < 0.05$), respectively.

6. Conclusions and Future Research

Building upon the comprehensive data analysis provided previously, this section will be segmented into two pivotal areas: conclusion and future research.

6.1. Conclusions

To improve user satisfaction with emergency rescue drone products, a product design method based on the fuzzy Kano-QFD-FBS model is proposed. The fuzzy Kano model is used to filter out key user requirements, the QFD model is applied to construct a house of high quality to obtain design elements and weights for emergency rescue drone products, and the mapping transformation from functional elements to structural elements of emergency rescue drone products through the FBS model is completed to achieve the shape design of emergency rescue drone products. The user satisfaction evaluation of the emergency rescue drone design scheme is conducted through the use of a user satisfaction rating scale and a system usability scale based on expert review. The new design scheme can obtain higher user ratings compared to the original design, and the rating results have significant differences. Research shows that the product design method based on the fuzzy Kano-QFD-FBS model can comprehensively capture and analyse user requirements, accurately map them to product design elements, and produce specific plans for the design of emergency rescue drone products. The design scheme performs well in meeting user requirements, verifying the feasibility and effectiveness of the fuzzy Kano-QFD-FBS model in the research of drone shape design for emergency rescue, providing a new approach to product design for emergency rescue.

Hefei Jiaxun Technology Co., Ltd. carried out product development for the design scheme. At present, physical products have been sold on the market and have achieved good results. Hefei Jiaxun Technology Co., Ltd. conducted a survey on consumer satisfaction with this product, and the results revealed that customer satisfaction increased by 11.9% compared with that of previous products. Compared with similar products in the market, the consumer satisfaction with this product increased by 13.5%, indicating that it has obvious market competitiveness. The design scheme of emergency rescue drones improves the core competitiveness of products, solves the problem of product homogeneity, enhances user satisfaction, and creates huge value for enterprises. The development of the

emergency rescue drone design scheme has further implemented the cooperation between schools, enterprises and research institutes, broadened the scope of cooperation, and enhanced the connotation of cooperation.

However, in the user research stage of this method, the number of survey questionnaires collected in some areas was small, which may lead to information deviation, affect the accuracy of demand acquisition, and thus affect the later design. The product design practice stage involves sensor technology and communication transmission technology. Only preliminary feasibility was considered, without in-depth analysis of specific configurations and usages. The specific parameters remain at the feasibility level.

6.2. Future Studies

In future research, improvements can focus on a key area of the user requirements system. First, in the early stages of product design, the selection and construction of user requirements are overly subjective. For the screening of user requirements, the first step is to investigate the product through a literature inquiry, a questionnaire survey, and a user interview to obtain a description of the user requirements for the product and then select specific user requirement evaluation items through the expert opinion method. However, in many real-world situations, the expert opinion method cannot be used to accurately interpret quantitative values. In subsequent design research, we will quantitatively screen user requirements to ensure the scientific and rational nature of the user requirements project system. Second, in the later stages of product design, there is a low match between user requirements and design solutions, resulting in a lack of scientific and quantitative design methods. In the future, more systematic and in-depth research can be conducted in combination with product shape semantics.

Author Contributions: Conceptualization, Z.W. and K.Y.; methodology, Z.W.; validation, Z.W.; formal analysis, Z.W.; investigation, Z.W. and K.Y.; resources, Z.Z.; data curation, X.L.; writing—review and editing, Z.W.; visualization, Y.W.; supervision, Y.W.; project administration, Z.W.; funding acquisition, Z.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the project of Anhui Provincial Philosophy and Social Science Foundation Planning Project, grant number AHSKQ2022D154.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data are available from the authors upon request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Yu, Q.; Xing, G. Application and development of UAV in emergency rescue in our country. *Occup. Health Emerg. Rescue* **2021**, *39*, 350–355.
2. Nan, J.L. The research and application prospect analysis of fire UAV. *Fire Sci. Technol.* **2017**, *36*, 1105–1107+1112.
3. Chen, G. Explore the fire-fighting control system and key technologies of unmanned aerial vehicles. *China Plant Eng.* **2023**, *3*, 28–30.
4. Wang, J.; Zhang, L.; Wu, C.Y. Research on visual algorithm of unmanned aerial vehicle in high-rise building fire fighting. *Fire Sci. Technol.* **2018**, *37*, 1704–1706.
5. Yin, X.Y.; Chang, Y.; Liu, B.S. Modeling Design of Detection UAV Based on SGS Model and AHP. *Packag. Eng.* **2022**, *43*, 45–50.
6. Qi, X.Q.; Zhang, Q.; Han, J.S.; Xue, S.J.; Wang, W.J.; He, C. Design of Tourism Service Drone Based on User Experience Optimization. *Packag. Eng.* **2020**, *41*, 83–89.

7. Qin, Y.X.; Chen, Y.; Qiao, H.H.; Chen, Z.Q.; Zhang, G.P. Design of Dynamic Track Planning Algorithm for Disaster Detection UAV. *Infrared Laser Eng.* **2019**, *48*, 284–289.
8. Gao, J.J.; Yu, S.H.; Zhao, X.D. Multi-hierarchy Grey Relation Evaluation Method on UAV Styling Design. *Packag. Eng.* **2020**, *41*, 106–110.
9. Sheng, W.; Lu, D.H. Design and implementation of cloud-based UAV remote technical support system. Comput. *Eng. Des.* **2021**, *41*, 1787–1795.
10. Wang, D.S.; Liu, X.G.; Li, W.; Zhang, J.X.; Yuan, T.; Zhang, C.L. Precision Autonomous Flight Control Method of UAV Based on Multi-sensor Integration. *Trans. Chin. Soc. Agric. Mach.* **2019**, *50*, 98–106.
11. Zhen, X.Z.; Tong, X.; Zhang, Y.; Guo, J.; Guo, Y.N. Discussion on key technologies of multirotor detection UAVs in mine dangerous area. *Ind. Mine Autom.* **2020**, *46*, 48–56.
12. Zhang, X.X.; Yan, B.X.; Sun GK Song, Y.M. The Design of Flexible Deformation Wing and Structural Mechanical Property Analysis. *Mach. Des. Manuf.* **2020**, *5*, 163–166.
13. Wang, W.L.; Ma, M.X.; Hu, Y.Z.; Zhu, J.F. Application of TRIZ Theory in Design of Launch Recovery Device for UAV Cluster. *Mach. Des. Manuf.* **2020**, *12*, 108–112.
14. Wang, X.H.; Meng, X.Y.; Li, C.X. Design of trajectory tracking controller for UAV based on MPC. *Syst. Eng. Electron.* **2021**, *43*, 191–198.
15. Qin, C.J.; Li, Y. Research on Automatic Identification Technology of Crop Diseases and Insect Pests: Based on Video Monitoring and UAV Platform. *J. Agric. Mech. Res.* **2021**, *43*, 42–45.
16. Shan, C.Y.; Yang, W.; Geng, C.B. Pose Estimation Method Based on Artificial Landmarks for the Autonomous Flight of Underground UAVs. *J. China Coal Soc.* **2019**, *44* (Suppl. S1), 387–398.
17. Chen, M.C.; Hsu CL Lee, L.H. Service quality and customer satisfaction in pharmaceutical logistics: An analysis based on Kano model and importance-satisfaction model. *Int. J. Environ. Res. Public Health* **2019**, *21*, 4091.
18. Wu, J.; Hei, Y.; Fan, Y.; Zhou, X. An Analysis of the Requirements for Smart Guiding Services in Museums Using the Kano-AHP Method. *J. Artif. Intell. Pract.* **2023**, *6*, 45–57.
19. Dou, R.; Zhang, Y.; Nan, G. Application of combined Kano model and interactive genetic algorithm for product customization. *J. Intell. Manuf.* **2019**, *30*, 2587–2602.
20. Noriaki, K.; Seraku, N.; Takahashi, F.; Tsuji, S. Attractive quality and must-be quality. *J. Jpn. Soc. Qual. Control* **1984**, *14*, 39–48.
21. Shravan, M.B.; Manjunath, S.; Abhiram, M. Kano Model for Enhancing Business Strategy through Customer Preference Analysis: A Case Study of Packaged Mineral Water Bottles. *IEEE Eng. Manag. Rev.* **2024**, *52*, 172–185.
22. Wang, Z.Y. Research and Application of Product Design User Requirements Mining Based on Online Comments and Kano Model. *Informatica* **2024**, *47*, 141–154.
23. Chen, X.; Qiu, D.P. Research of Product Design Based on Fuzzy-Kano Model and TOPSIS. *J. Graph.* **2019**, *40*, 315–320.
24. Zhang, W.X.; Xiao, R.B.; Lin, W.G. Review Data Driven Customer Need Model Research Based on Product Performance Lexicon. *China Mech. Eng.* **2020**, *31*, 1866–1876.
25. Hao, C.J.; He, R.K.; Yin, L.; Ma, K. Fault Expression of Home Robot Vacuum Based on Emotional Interaction. *Packag. Eng.* **2022**, *43*, 98–105.
26. Wang, T.Q. Design and research of smart air purifier based on fuzzy Kano model. *J. Mach. Des.* **2017**, *34*, 122–125.
27. Li, H. Research on Human-Machine Interface Design of Domestic Treadmill Based on Kano-QFD. Master's Thesis, Taiyuan University of Technology, Taiyuan, China, 2021.
28. Akao, Y. New product development and quality assurance–quality deployment system. *Stand. Qual. Control* **1972**, *25*, 7–14.
29. Su, C.; Zhang, S.L.; Xiao, L. Modeling Design of Corrugated Board Production Line Based on QFD and TRIZ. *Packag. Eng.* **2019**, *40*, 118–124.
30. Lv, Z.Y.; Yang, B.; Huang, F. Appearance Design and Improvement of Complex Product Based on QFD. *Mech. Des.* **2019**, *36*, 119–126.
31. Yu, M.W.; Wang, L.Z. Improvement of Health Examination Service Quality Based on QFD. *China Health Serv. Manag.* **2018**, *35*, 465–468.
32. Xu, Y.Q.; Song, D.S.; Gu, J.L.; Feng, W.D. Multiadaptive Design of Toilet Based on QFD and FBS Model. *Packag. Eng.* **2018**, *39*, 283–287.
33. Zhou, J.J.; Fan, J.J.; Shi, Y.W.; Chen, W.H. Research on Development and Design of Medical Disinfection Products Based on QFD and DOE Model. *Mach. Des. Res.* **2020**, *36*, 205–209.

34. Luo, X.G.; Zhao, Z.X.; Li, J.; Zhang, Z.L. Literature review and prospect analysis for quantitative models and methods of quality funetion deployment. *Comput. Integr. Manuf. Syst.* **2023**, *29*, 1668–1683.
35. Avikal, S.; Singh, R.; Rashmi, R. QFD and Fuzzy Kano model based approach for classification of aesthetic attributes of SUV car profile. *J. Intell. Manuf.* **2020**, *31*, 271–284.
36. Cascini, G.; Fantoni, G.; Montagna, F. Situating needs and requirements in the FBS framework. *Des. Stud.* **2013**, *34*, 636–662.
37. Gero, J.S.; Kannengiesser, U. The situated Function-Behaviour-Structure framework. *Des. Stud.* **2004**, *25*, 373–391.
38. Wang, W.W.; Zhang, Y.; Jin, Q.Y.; Song, J.L. Study on the Innovative Furniture Design Method Based on Situational Analysis and FBS Mode. *Furnit. Inter. Des.* **2022**, *29*, 42–47.
39. Zhou, Q.; Li, X.; Zhou, J.Y. Integrated innovative design method of fuzzy Kano and scenario FBS model. *J. Graph.* **2020**, *5*, 796–804.
40. Chen, G.Q.; Dai, C.; Shen, Z.Y.; Zhang, P.; Tang, Z.C. Design of Movable Electric Testing Station Based on QFD and FBS. *Packag. Eng.* **2021**, *42*, 43–50.
41. Büyüközkan, G.; Güler, M.; Mukul, E. An integrated fuzzy QFD methodology for customer oriented multifunctional power bank design. *Cust. Oriented Prod. Des. Intell. Fuzzy Tech.* **2020**, *279*, 73–91.
42. Wu, X.; Hong, Z.; Li, Y.; Zhou, F.; Niu, Y.; Xue, C. A function combined baby stroller design method developed by fusing Kano, QFD and FAST methodologies. *Int. J. Ind. Ergon.* **2020**, *75*, 102867.
43. Rampal, A.; Mehra, A.; Singh, R.; Yadav, A.; Nath, K.; Chauhan, A.S. Kano and QFD analyses for autonomous electric car: Design for enhancing customer contentment. *Mater. Today Proc.* **2022**, *62*, 1481–1488.
44. Chen, C.; Zhang, C.; Xu, Z. An Online Reviews-Driven Kano-QFD Method for Service Design. *IEEE Trans. Eng. Manag.* **2024**, *71*, 8153–8165.
45. Li, J.; Kim, K. Kano-QFD-based analysis of the influence of user experience on the design of handicraft intangible cultural heritage apps. *Herit. Sci.* **2023**, *11*, 59.
46. Zhao, D.; Chen, W. Design and Research of Smart Neck Helmets Based on the KANO-QFDModel and TRIZ Theory. *Secur. Commun. Netw.* **2021**, *1*, 4693719.
47. Tandiono, Y.; Rau, H. An enhanced model using the Kano Model, QFDE, and TRIZ with a component-based approach for sustainable and innovative product design. *Sustainability* **2022**, *15*, 527.
48. Wan, H.; Zhao, W.; Li, W.; Chen, L.; Guo, X. An innovative device to transport electron gun automatically in nuclear power plant. *J. Nucl. Sci. Technol.* **2020**, *57*, 40–48.
49. Wang, J.; Liu, X.T.; Liu, J.; Yan, J.X.; Xu, Y.F. Innovative Design of Children's Swimming Auxiliary Products Based on QFD-FBS. *Packag. Eng.* **2024**, *45*, 200–206.
50. Li, Y.X.; Zhu, Y.X.; Wo, C.W.; Zheng, L.X.; Chen, M. Design of Braille Writing Modifier Based on QFD and FBS Model. *Packag. Eng.* **2023**, *44*, 185–192.
51. Su, C.; Tang, P.; Song, R.X.; Luo, G.F. Design of Smart Bed Based on Demand Analysis and Situational FBS Model. *Mach. Des. Res.* **2021**, *37*, 7–12+17.
52. Chaudha, A.; Jain, R.; Singh, A.R.; Mishra, P.K. Integration of kano's model into quality function deployment (qfd*). Int. J. Adv. Manuf. Technol.* **2011**, *53*, 689–698.
53. Kaya, A.; Ozturk, R.; Altin Gumussoy, C. Usability measurement of mobile applications with system usability scale (SUS). In Proceedings of the Industrial Engineering in the Big Data Era: Selected Papers from the Global Joint Conference on Industrial Engineering and Its Application Areas, GJCIE 2018, Nevsehir, Turkey, 21–22 June 2018; pp. 389–400.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.